

Natural Language Technology

Week 2 - Language Input Technology

Language Input Technologies



Today

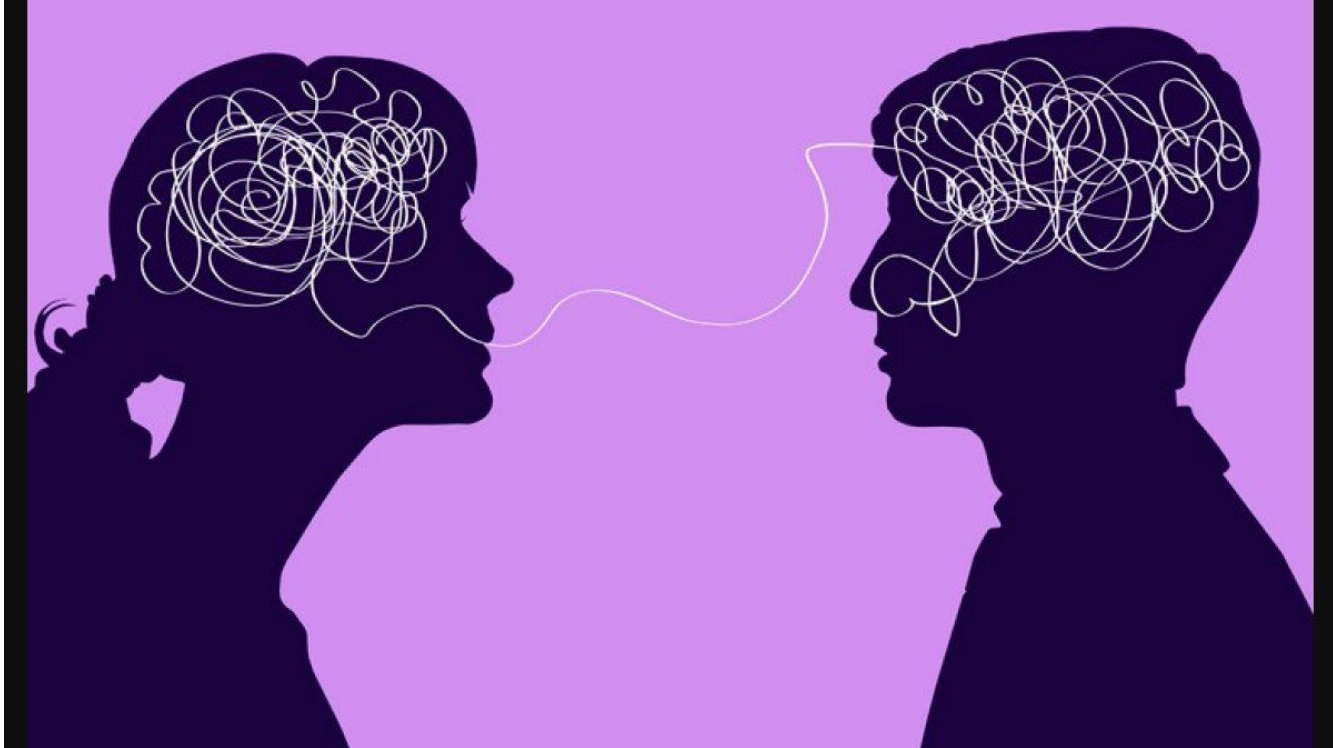
- What is Language?
- What are the properties of Language which have made it challenging to process?
- Optical Character Recognition
- Handwriting Recognition
- Speech Recognition
- Speaker Recognition

What is Language?

- Definitions:
 - A system of communication used by a particular country or community.
 - The principal method of human communication, **consisting of words** used in a structured and conventional way and conveyed by speech, writing, or gesture.



Why is Language Tough?



A slow serialized form of communication – requires a lot of context!

Language is Overloaded

- We often use the same sounds and or words for different meanings.
 - Homonym



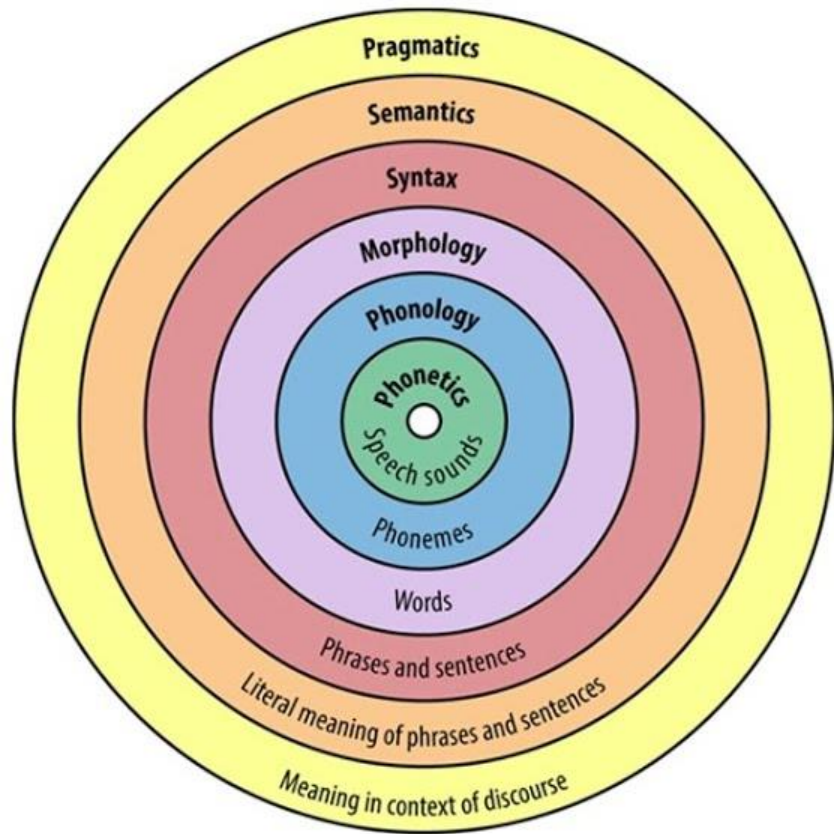
Credit: <https://www.twinkl.de/teaching-wiki/homonym>

- We often use the different sounds and words for the same meanings.
 - Synonyms
 - How many real synonyms can you think of?

Language Mechanisms and Challenges

- To overcome its inherent limits, language evolved to make use of some important features / tricks
- Anaphoric reference – referring back to something mentioned earlier, e.g., **pronoun use**
 - Cataphoric reference – referring forward
 - Exophoric reference – referring to something outside of the text
- Ellipsis
 - The omission of words from speech or text from an expression that can be instead inferred from the context
 - Verbal ellipsis: **If I could have made the dinner, I would have (...)**

Level Of Linguistics



- **Pragmatics:** study of language in communication
- **Semantics:** study of meanings in sentence.
- **Syntax:** study of sentence structure.
- **Morphology:** study of word structure.
- **Phonology:** study of arrangement of sounds.
- **Phonetics:** study of production of sounds.

Important Terminology

- Please see the uploaded PDF for important definitions

Linguistics and Natural Language Technology

"Every time I fire a linguist, the performance of the speech recognizer goes up"



Frederick Jelinek, IBM

- Linguistics does give us a good common set of terminology to refer to the things and processes that we are trying to replicate and make better – even if the methods that we use don't always have to fully take advantage of those things.

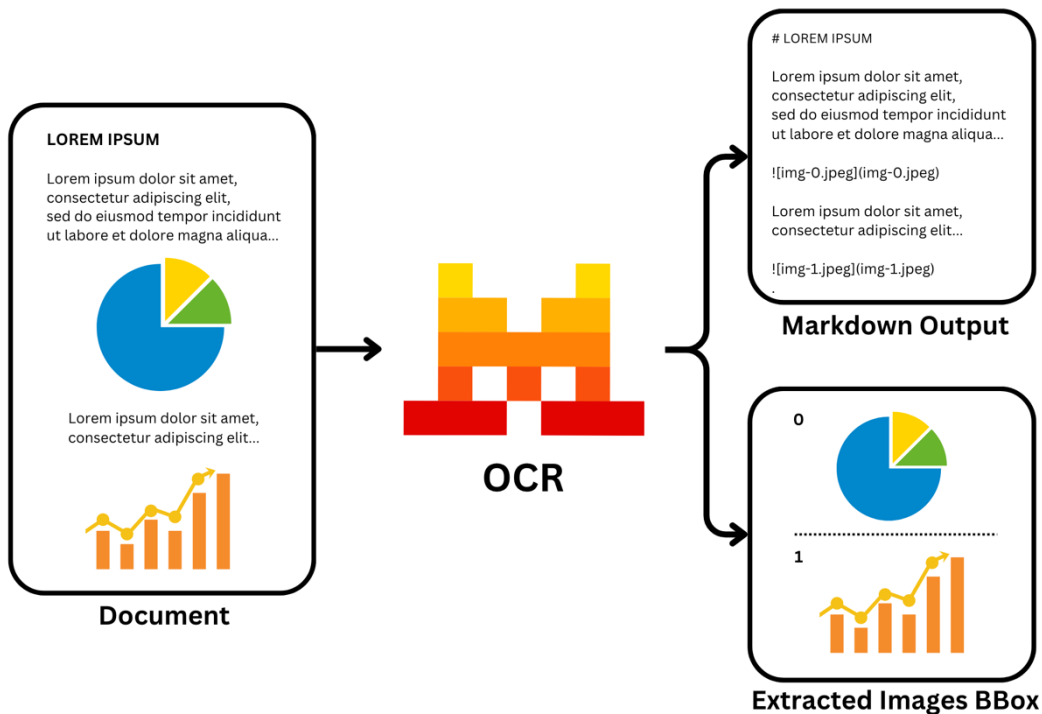
Some Input Types

Optical Character Recognition

- Input Data = Written Text
 - Historically pages of books / documents
 - Now also signposts, menus etc.
- Challenges
 - Lighting, contrast, print quality, font style etc.
 - Skewing of source data
 - Document Complexity
- State of the art
 - Amazon Textract
 - Google Document AI
 - Open Source: mmocr



‘LLMs’ adding to the mix.... Mistral OCR



Optical Character Recognition

- Word Error Rate: Proportion of incorrect words read

- where

- S = substitutions (number of incorrect words)
- D = number of words missed
- I = Number of words inserted incorrectly
- C = Correctly identified words

$$WER = \frac{S + D + I}{N} = \frac{S + D + I}{S + D + C}$$

Calculate for:

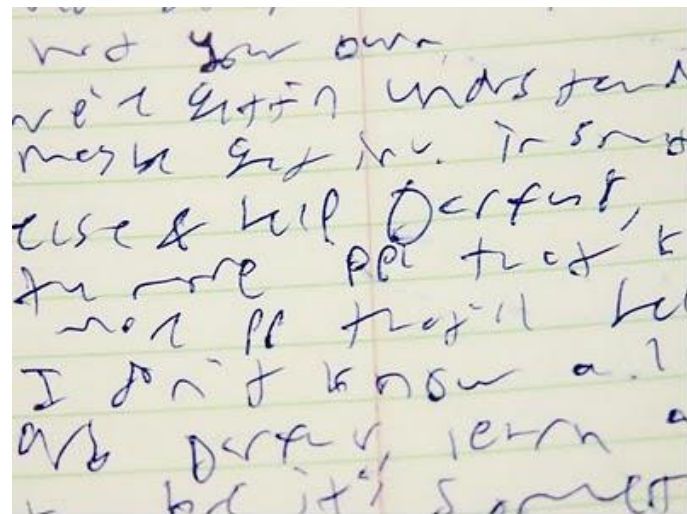
Ref = "The cat sat on the mat"

Hypothesis = "the bat sat on the fat mat"

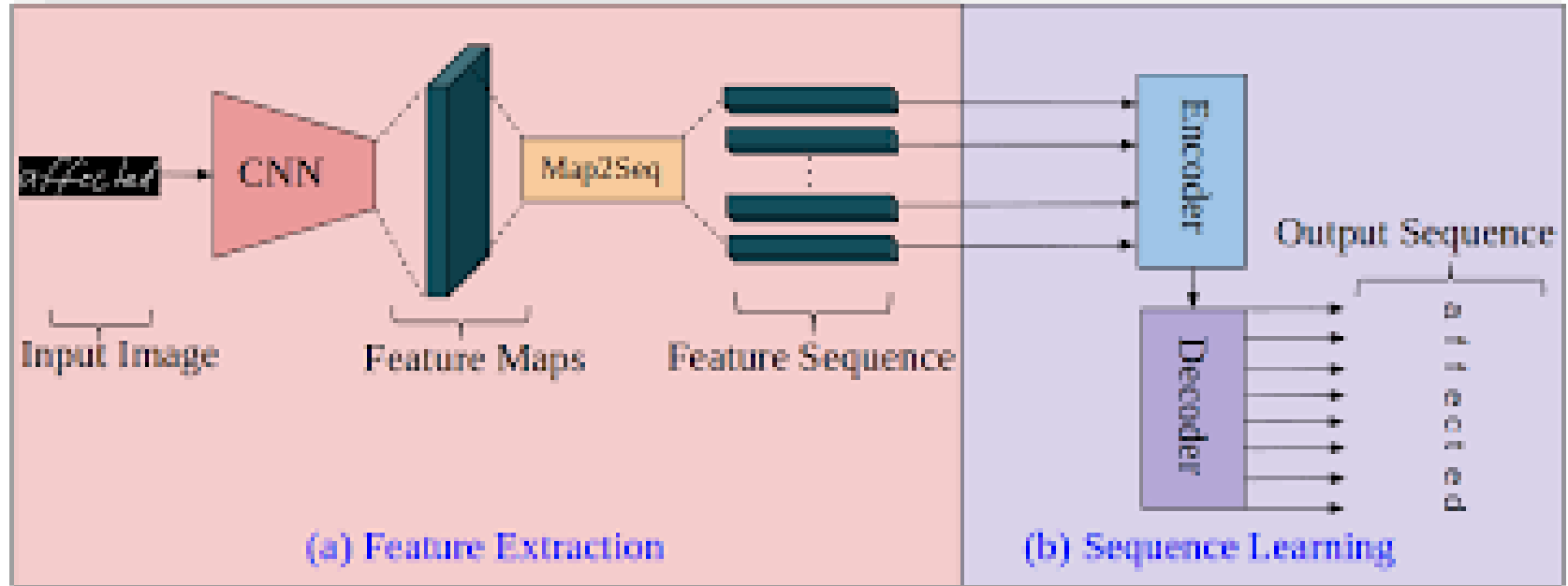
- Consider also Character Error Rate

Handwriting Recognition

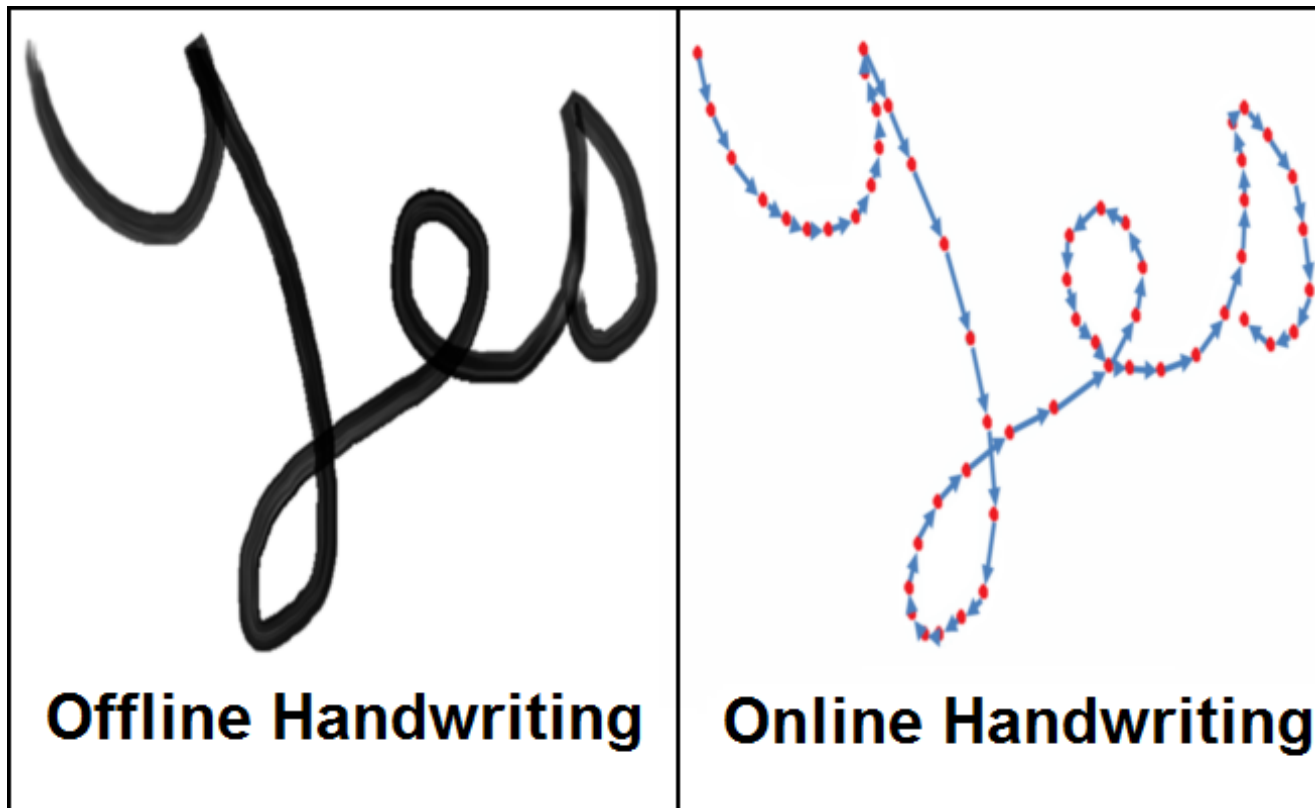
- Input data = handwritten text (offline or online)
 - Offline = image
 - Online = information on pen position over time
 - Terminology is a bit confusing as overlap with idea of synchronous vs asynchronous processing
- Challenges:
 - See OCR
 - Messy text / handwriting styles
 - Incorrect spelling
 - Segmenting text
- State of the Art
 - MyScript Handwriting Technology (Nebo)
 - Gboard (google)
 - Open: Tesseract (Google) TrOCR (Microsoft)



Offline Handwriting Recognition (Modern Model)

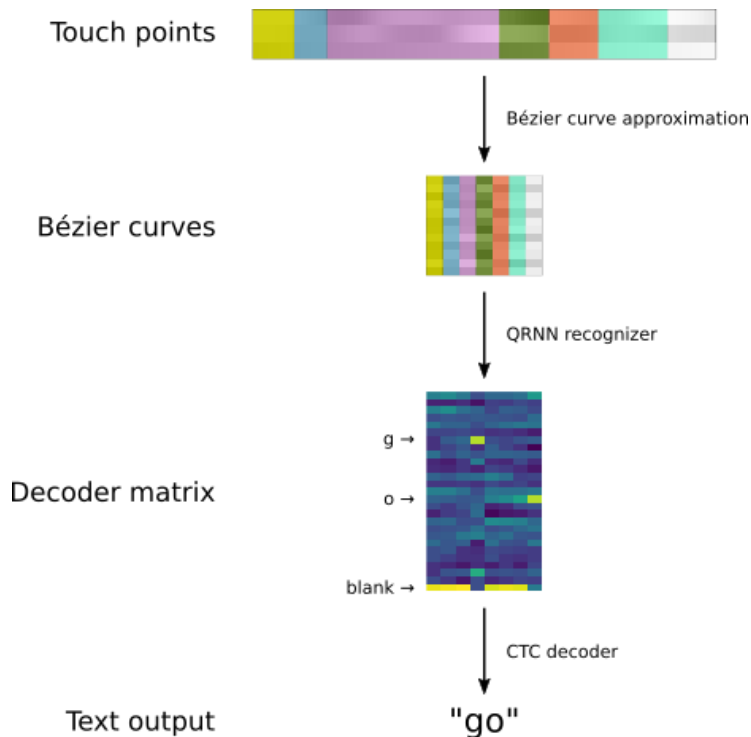


Online Handwriting Recognition



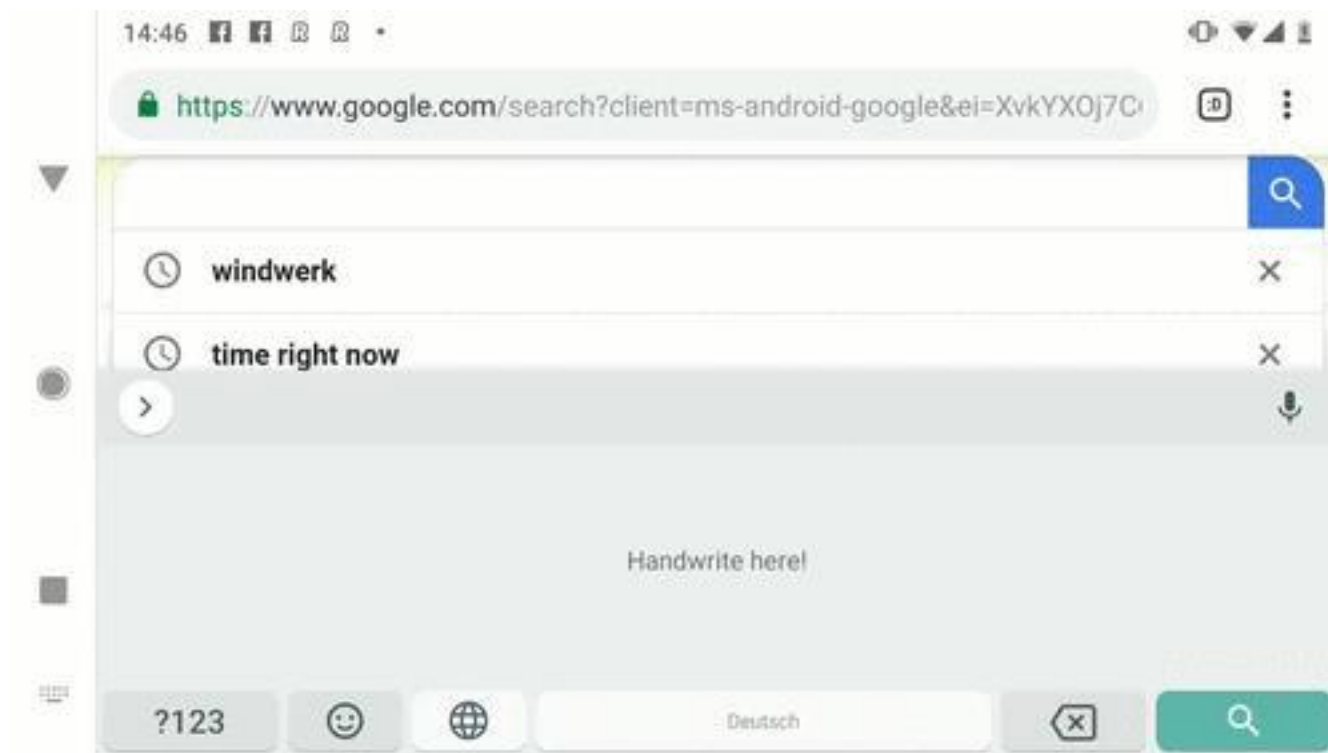
An image vs a sequence of small actions.

Online Handwriting Recognition



The details of a model like this are not important to us here, but it is important to understand the difference between processing the input as an image versus a series of touch points

Online Handwriting Recognition



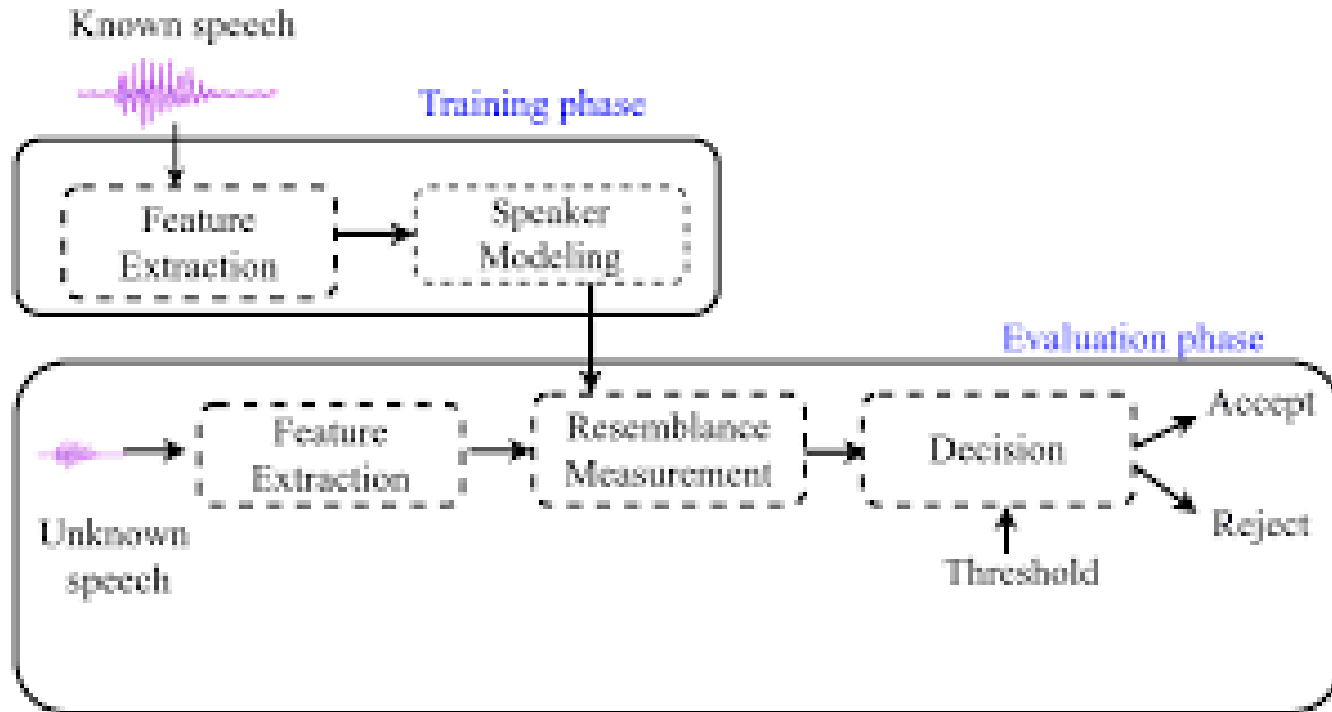
<https://ai.googleblog.com/2019/03/rnn-based-handwriting-recognition-in.html>

Speaker Recognition

- Input Data = Audio Sample / Extract
 - Output = Speaker ID typically with likelihood
- Speaker Verification vs Speaker Identification
 - Speaker Verification = 1:1 process focused on security
 - Speaker Identification = Determining most likely speaker from group (1:n)
- State of the Art
 - Google Cloud Speaker ID
 - Amazon Connect Voice ID
 - Open Source = SpeechBrain (does a lot more as well)



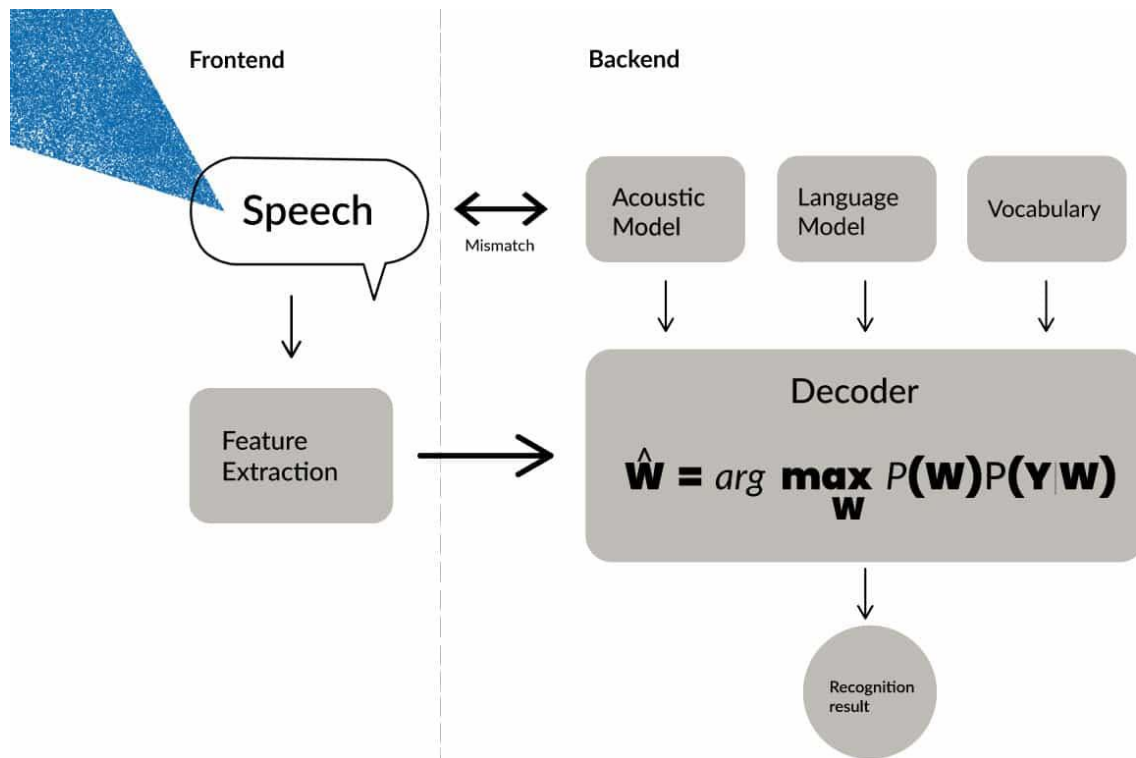
Speaker Recognition (Generic Architecture)



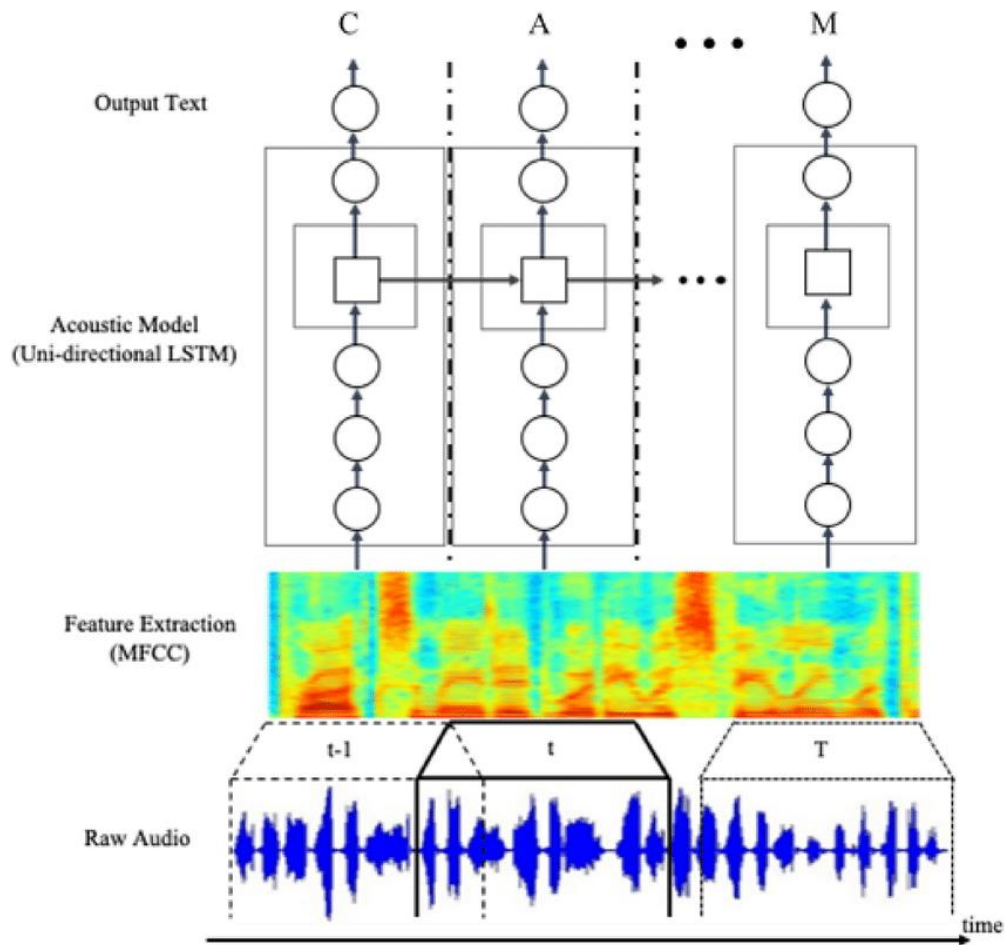
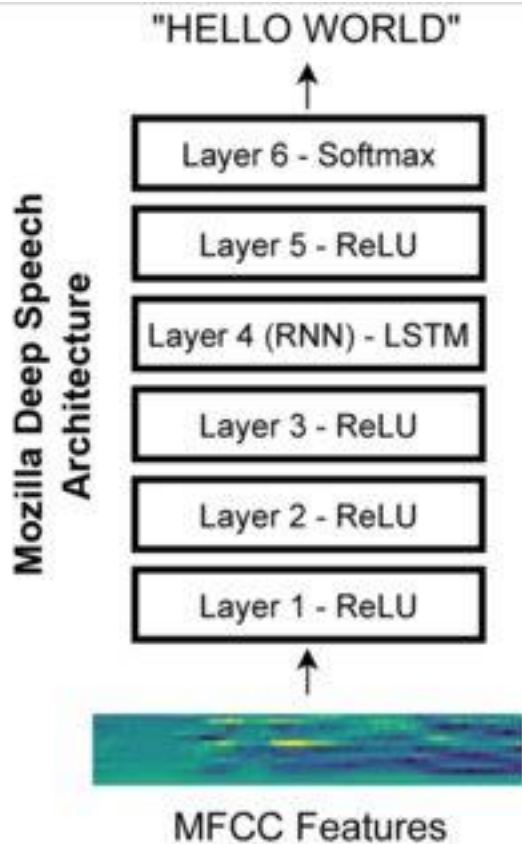
Speech Recognition

- Input Data = Audio Sample / Extract / Live
 - Output = the most likely word sequence, usually with timings and confidence
- Long studied challenge – and the one that has moved fastest since 2020
- State of the Art (2026)
 - Open weights: NVIDIA Parakeet / Canary, Alibaba Qwen3-ASR, IBM Granite Speech, OpenAI Whisper
 - Commercial APIs: Deepgram Nova-3, OpenAI gpt-4o-transcribe, ElevenLabs Scribe v2, Google Chirp 3
 - Open weights now match or beat the paid APIs on English

Classic Speech Recognition Model



Deep Speech Recognition



Typical Deep Speech Recognition

- Transcribed Aligned Data (expensive)
- Audio mapped to Acoustic Features (time series)
- A machine learning algorithm capable of processing time series / sequential data then processes the content.
- The content produces at any given point the most likely 'word / symbol' given (a) the acoustic inputs, and (b) the language context.
 - Acoustic input is travelling upwards is prior diagram
 - Language context is travelling left to right
- The acoustic model is captured in the network weights

From DeepSpeech to 2026 – Three Things Changed

- The encoder stopped being an RNN
 - Conformer = convolution for local phonetic detail + self-attention for context
- Supervision got cheap
 - Whisper: 680k hours of weakly supervised web audio beats curated corpora for robustness
 - Self-supervised pretraining (wav2vec 2.0, HuBERT) needs no transcripts at all
- The decoder became a language model
 - The 'language context' arrow on the previous slides is now literally an LLM
- We did not just fire the linguists – we fired the lexicon, the pronunciation dictionary and the separate acoustic model too

Backbones: Choice of Encoder

Encoder	Idea	Used by
Conformer / FastConformer	Convolution blocks for local spectral detail, self-attention for long-range context, heavy subsampling to ~80 ms frames	Parakeet, Canary, Nemotron
Whisper-style Transformer	Plain transformer over 30 s log-Mel windows, trained supervised at web scale	Whisper, Distil-Whisper
Self-supervised (wav2vec 2.0, HuBERT)	Contrastive / masked prediction on unlabelled audio, then fine-tune on a few hours	Meta Omnilingual ASR, most clinical work
Causal / streaming	Same blocks, attention restricted so no right context is needed	Moonshine v2, Kyutai STT, Voxtral Realtime

Every current system = an encoder (this slide) × an alignment strategy (next slide).

Open ASR Leaderboard – September 2026

Model	Backbone	WER	RTFx	Licence
Granite Speech 4.1 2B (IBM)	Speech-LLM	5.33	231	Apache 2.0
Cohere Labs Transcribe	Conformer + Transformer	5.42	525	Apache 2.0
Canary-Qwen-2.5B (NVIDIA)	FastConformer + Qwen3	5.63	418	CC-BY-4.0
Qwen3-ASR-1.7B (Alibaba)	Speech-LLM	5.76	–	Apache 2.0
Parakeet-TDT-0.6B-v3 (NVIDIA)	FastConformer + TDT	6.34	3333	CC-BY-4.0
Whisper large-v3 (OpenAI)	Attention enc-dec	~7.4	~40	MIT
Parakeet-CTC-0.6B (NVIDIA)	FastConformer + CTC	8.96	6400	CC-BY-4.0

Mean WER over LibriSpeech, GigaSpeech, AMI, Earnings22, SPGISpeech, TED-LIUM and VoxPopuli, after text normalisation. RTFx = seconds of audio transcribed per second of compute.

Reading the Leaderboard Critically

- Whisper is no longer state of the art on English
 - About ten open models beat it – but it is still the best documented, most permissively licensed and widest-language default
- The accuracy spread across the top ten is about one WER point
 - The speed spread is about thirty-fold – architecture choice in 2026 is a latency decision, not an accuracy one
- Re-benchmarking on task audio advised before choosing

Beyond English

- Parakeet-TDT-0.6B-v3 – the practical multilingual workhorse
 - 25 European languages, automatic language detection, 0.6B parameters, CC-BY-4.0
- Meta Omnilingual ASR (Nov 2025) – what self-supervision bought us
 - wav2vec 2.0 encoders from 300M to 7B parameters
 - 1,600+ languages natively; 5,400+ zero-shot by in-context learning from a handful of audio–text pairs
 - Character error rate below 10% for 78% of those languages
- Irish, code-switching and low-resource speech are now a fine-tuning problem, not a data-collection moonshot

Measuring ASR in 2026 – WER Is Not Enough

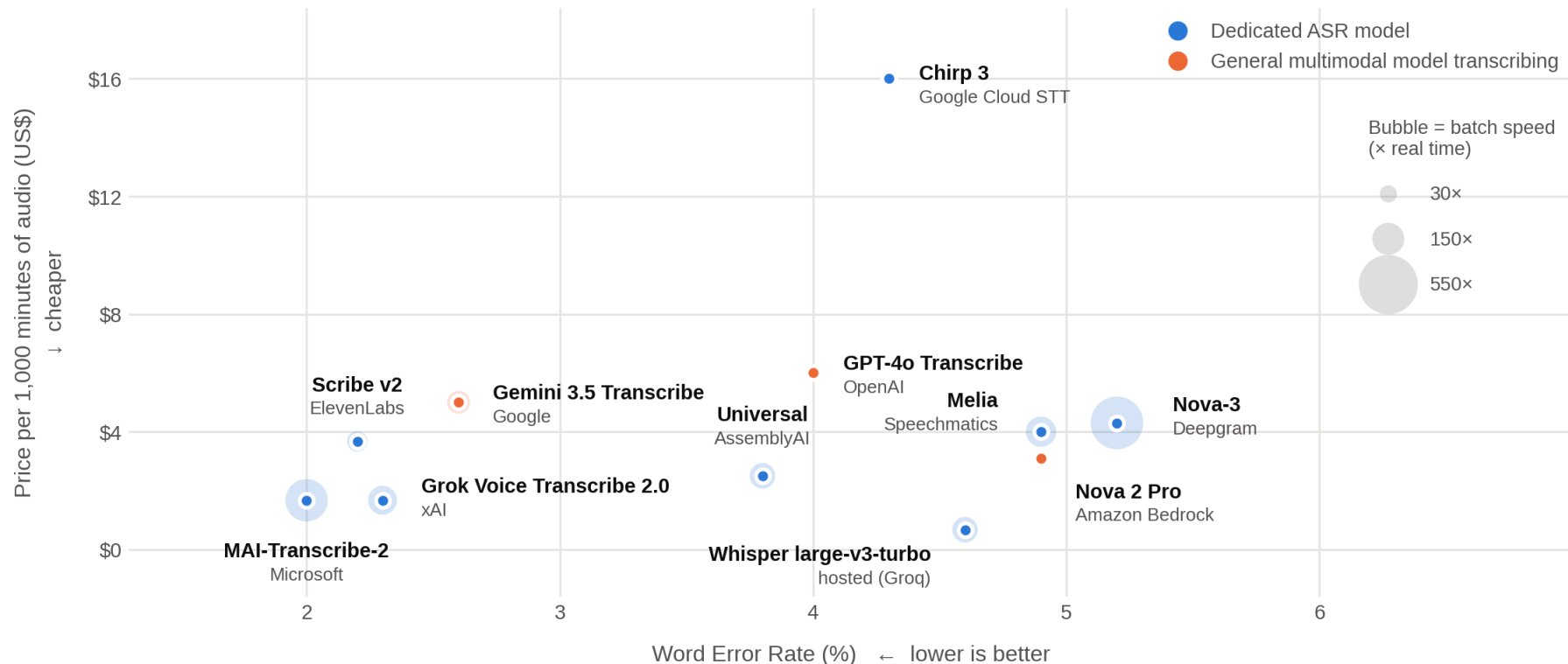
- Character Error Rate for low-resource and disordered speech
- Entity or keyword error rate
 - Getting 'Aoife' and 'warfarin' right matters more than the function words that dominate WER
- Speed is three different numbers
 - RTFx (batch throughput) $\neq$ streaming latency $\neq$ end-of-turn latency
- Report WER per accent, per age group, per speaker – aggregate WER hides the failures that matter

Where the Frontier Actually Is

- Clean, read, adult, native-accented English is solved to within annotator disagreement
- Still open, and all of them make good projects:
 - Atypical and disordered speech – dysarthric ASR still runs multiples worse; the live work is parameter-efficient adaptation of foundation models (LoRA on Whisper, personalised fine-tuning)
 - Overlapping speech and diarisation in real meetings
 - Code-switching, including Irish–English
 - Children’s speech – scarce data, difficult consent
 - Quality at low latency – the gap between batch WER and 150 ms-latency WER is still real

Commercial speech recognition: accuracy vs cost

September 2026 · English batch transcription



Word error rate: Artificial Analysis AA-WER v2 (AA-AgentTalk 50%, VoxPopuli 25%, Earnings22 25%, Whisper-normalised). Price is list price for batch English. Open-weight models (Parakeet-TDT, Canary-Qwen, Whisper) sit off this chart: no licence cost at all, you pay only for the GPU you run them on.

Week 2 – Lab work

- Record audio of yourself in a Colab notebook – no accounts, no API keys, no billing
- Transcribe it with two generations of architecture
- Calculate Word Error Rate for each, before and after text normalisation
- Sweep the signal-to-noise ratio and plot WER per model – compare how they degrade
- Feed each model ten seconds of silence and record what it invents
- Notebook: `NLT_Week2_ASR_Lab_2026.ipynb`